

Andrew Coggins

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EDUCATION

Texas A&M University College Station, Texas

Bachelor of Science in Electrical and Computer Engineering
minors in Mathematics and AI in Business

Graduation May 2028

Cumulative GPA: 3.5/4.00

University GPA: 3.4/4.00

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, Verilog HDL, ARMv8 Assembly

Developer Tools & Hardware: Git, GitHub, VS Code, Linux/Unix, Claude Code, Xilinx Vivado, LTspice, IVerilog, GTKWave

Relevant Coursework: Data Structures & Algorithms, Program Design & Concepts, Discrete Structures, Advanced Computer Architecture & Design, Digital System Design, Electrical Circuit Theory, Differential Equations

PROJECTS

Class Schedule Optimizer | Python, REST APIs, Web Scraping, Claude Code

- Built a data pipeline ingesting **21.6k+ course sections**, combining RateMyProfessors and grade archives (Anex.us) via fuzzy string matching.
- Developed weighted scoring and cart-syncing logic to evaluate GPA distributions and push to schedules directly

Image Processing & Scaling Engine | C++, Data Structures, Algorithmic Scaling, File I/O

- Built a C++ application using bicubic interpolation to perform spatial transformations on PPM image files.
- Constructed a 2D vector matrices in column-major order with index arithmetic to traverse 4x4 pixel matrices.

Single Cycle LEGv8 Processor | Verilog HDL, Computer Architecture, IVerilog, ARMv8 Assembly

- Designed and verified a **single-cycle ARMv8 RISC processor** in Verilog HDL, implementing a 64-bit ALU, multi-format Sign Extender, control unit, and branch-resolving NextPC logic. Included self-validation through comprehensive testing.

Autonomous Web Studio Agent | Claude Code, Vercel, Git

- Authored custom Claude Code agent skills to orchestrate multi-step autonomous workflows, automating client research, prompt-driven content synthesis, and site layout generation.
- Architected an agentic build pipeline mapping structured AI outputs into templates for automated Vercel deployment

LEADERSHIP & INVOLVEMENT

One Army, Texas Aggie Men United | Class Candidate Advisor

September 2025 – Present

- One of eight leaders in charge of onboarding and mentoring new members for an organization of 90+ members
- Demonstrated leadership and project management skills within team as we marketed and executed part of the annual Gladiator Dash, a major fundraising event.
- In 2026, we raised a record **\$185,000**, a **2.75% increase** from last year.

Texas A&M Baja SAE | Business Lead

August 2026 – Present

- Directed cross-functional recruitment and operational management across accounting, sponsor relations, social media, financing, and operations sub teams while serving as primary liaison to Project Managers and university sponsors.
- Managed corporate partner relations, spearheading sponsorship acquisition, pitch execution, and industry stakeholder engagement to secure critical funding and material donations.

EMPLOYMENT AND INTERNATIONAL EXPERIENCE

Global Engineering Scholar | Queens University, Belfast, UKGB

Summer 2026

- Engaged with AMD R&D leaders on NPU/CPU/GPU compute stacks, performance benchmarking, and FPGA-accelerated quantized neural networks known as Xilinx FINN
- Evaluated embedded computer vision systems running object detection/face tracking algorithms on FPGA hardware at Queen's University Belfast. Observed semiconductor fabrication process including silicon wafer processing, transistor manufacturing, and MEMs device fabrication.

Private Tennis Instruction | Self employed

Texas, Summer 2024 - Present

- Built independent coaching practice from zero to 20+ active students using multi-channel business development